Online Retail EDA – Customer & Returns Analytics

Data-Driven Insights from 500,000 Transactions (2009–2011)

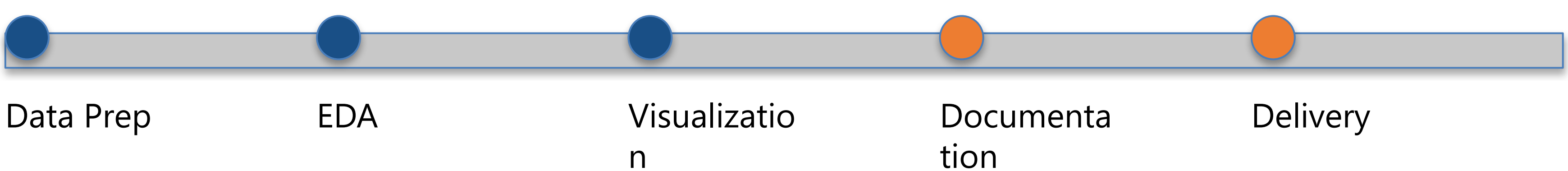
Artur Melnyk – Data Analyst

Python · Excel · Tableau

Project Overview

- Objective: Uncover revenue, customer, and return trends from Online Retail II dataset.
- Scope: Period 2009–2011 · ~500K transactions · 8 columns

Business Questions:
• What drives revenue growth?
• Who are top customers and high returners?
• Which products/countries contribute most?
Timeline: Data Prep → EDA → Visualization → Documentation → Delivery



Tools & Methodology

- Data Preparation – Python (Pandas, NumPy) – Handle nulls, clean types
- Exploration – Matplotlib, Seaborn – Identify sales & return trends
- Export for Viz – Excel – Prepare summary tables
- Visualization – Tableau – Build interactive dashboards
- Documentation – PDF & README – Ensure reproducibility

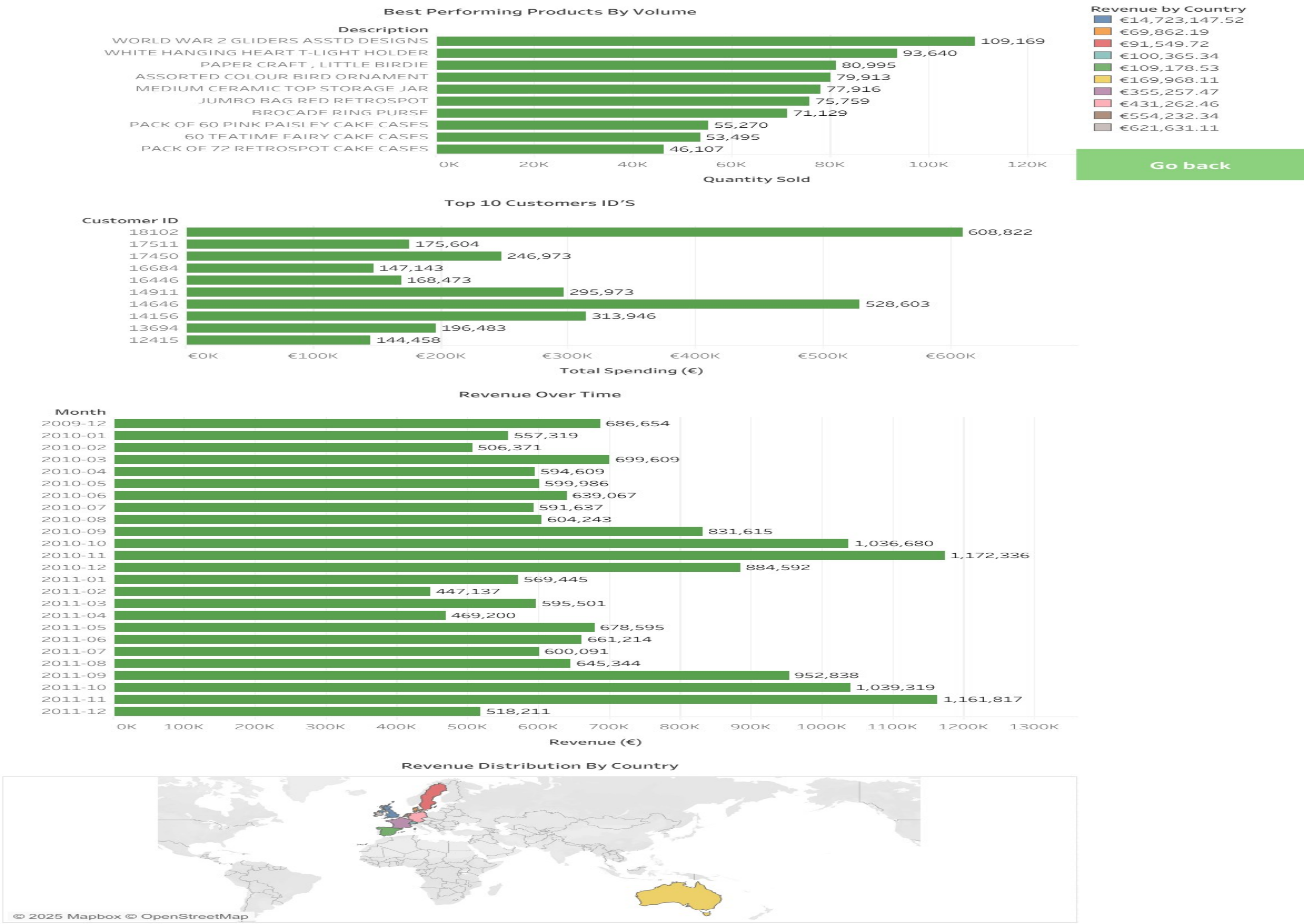
Data Cleaning Challenges & Solutions

- Mixed Date Formats – pd.to_datetime(errors='coerce')
- Missing Customer IDs – Dropped / flagged
- Currency Strings – Converted to float
- Invalid Descriptions – Flagged as Return Reasons
- Lesson: Data cleaning = foundation of credible analysis

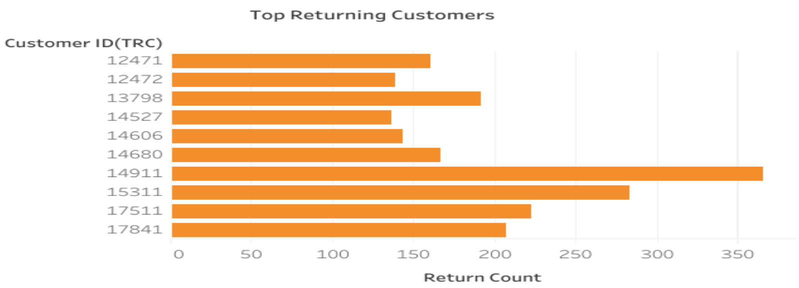
KPI Summary Bar

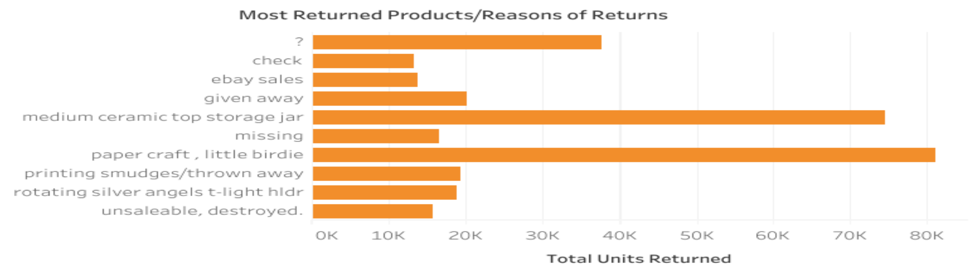
- 💶 Total Revenue: €14.7M (83% from UK)
- 👥 Top Customer (ID 18102): €608K spent
- 🔁 Average Return Rate: 7.3%
- 📦 Most Returned Product: Paper Craft Little Birdie
- 🌍 Countries Covered: 38 (UK & EIRE dominate)

Sales & Revenue Dashboard — Tableau Public



Customers & Returns Dashboard — Tableau





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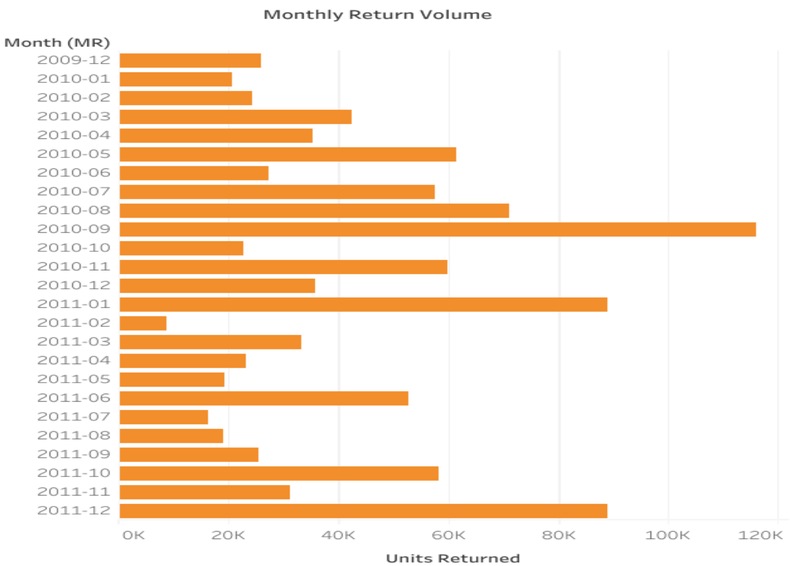
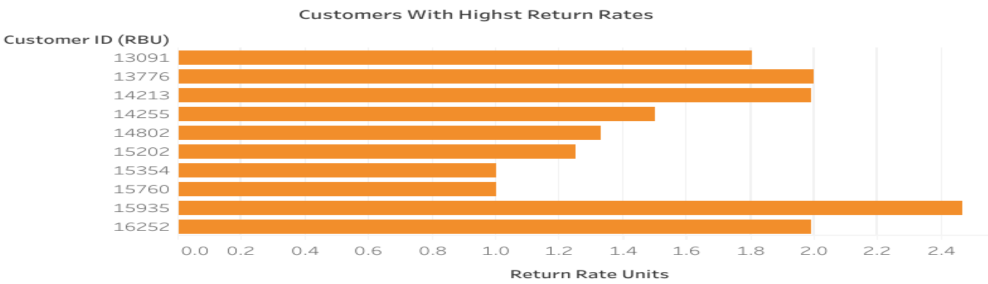
TOP RETURNING CUSTOMER
Customer ID: 14911
Returns: 366 Units

KPI - Total Units Returned

310,150





Top 3 Insights

1. GB UK accounts for >80% sales → Focus marketing/logistics on UK
2. 5% customers → 60% revenue → Loyalty & retention programs
3. Volume returners ≠ Rate returners → Differentiate return policies

Lessons Learned

1. Validate categorical semantics → detect hidden return reasons
2. Verify data types before Tableau import
3. Curate Top N → avoid clutter
4. Document continuously → reproducibility
5. Separate return metrics → volume vs rate clarity

Deliverables & Next Steps

- Deliverables: Notebook, Clean Excel, Tableau Dashboard, PDF Documentation, README
- Dashboard Link: Tableau Public (EDA Visualization)
- Next Steps: Extend to RFM segmentation, automate monthly return alerts

Summary & Closure

- Complete EDA case: raw data → insights → dashboards → documentation.
- Business-ready: actions tied to insights (UK focus, loyalty, differentiated returns).
- Reproducible workflow: notebook + cleaned dataset + Tableau + PDF + README.
- Executive communication: KPI slide, curated Top-N visuals, branded footer.
- Next: RFM segmentation, CLV model, automation of monthly return alerts.
- *From Data to Decisions — Artur Melnyk, 2025*